

Dihedral 3

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Claim 1a. For each $m > 0$ dividing n , let K_m be the set of solutions to $x^m = 1$, in μ_n . Show that K_m is a cyclic subgroup.

Proof. Let g be the generator of μ_n . Let $k \neq 0$ be the smallest integer s.t $a = g^k$ and $a^m = 1$ thus $a \in K_m$.

If there exists no $k \neq 0$ satisfying this condition, then $K_m = \{1\}$ or the trivial subgroup. Assume the contrary, there exists a $b \in \mu_n$ s.t $b^m = 1$ by definition $b = g^x$ for some $x \in [1, \dots, n-1]$ then $k = x$ which is a contradiction, thus $K_m = \{1\}$ if $k \nmid$.

If k exists. Then $\forall s \in \mathbb{Z}$, $g^{skm} = (g^m)^{sk} = 1^{sk} = 1$ thus $g^{sk} \in K_m$, meaning $\langle g^m \rangle \subseteq K_m$. $\forall a \in K_m$, since $a \in K_m \subset \mu_n$, $a = g^p$. By remainder theorem, $p = sk + r$ for $s \in \mathbb{Z}$ and $r \in [0, \dots, k-1]$. Thus $a = g^{ks+r}$. $a^m = 1 = (g^{ks+r})^m = g^{m(ks+r)} = g^{mks+mr} = g^{mks}g^r = g^{mr}$. Thus $(g^r)^m = 1$, however since k is by definition the smallest non zero integer s.t $(g^k)^m = 1$ and $r < k$ $r = 0$ because otherwise $r \neq 0$ and $r < k$ would be a contradiction. thus $a = sk + r = sk$. Thus $\forall a \in K_m$ $a \in \langle g^k \rangle$ thus $K_m \subset \langle g^k \rangle$. Thus since $K_m \subset \langle g^k \rangle$ and $\langle g^k \rangle \subset K_m$, $\langle g^k \rangle = K_m$ and K_m is cyclic. $\square$

Claim 1b. Show K_m has order m

Proof. Let g be the generator for μ_n . Since $n \mid n$ then $n/m \in \mathbb{Z}$. $\forall s \in \mathbb{Z}$, $(g^{\frac{n}{m}s})^m = g^{ns} = e$. Thus $\forall s \in \mathbb{Z}$, $g^{\frac{n}{m}s} \in K_m$ thus $\langle g^{\frac{n}{m}} \rangle \subseteq K_m$. Let $g^a \in K_m$, by definition $g^{am} = e = g^{sn}$ for $s \in \mathbb{Z}$. Then $am = sn$ and $a = sn/m$ thus $g^a = g^{s\frac{n}{m}}$ thus $g^a \in \langle g^{\frac{n}{m}} \rangle$. Thus $K_m \subset \langle g^{\frac{n}{m}} \rangle$ and $\langle g^{\frac{n}{m}} \rangle = K_m$. $\square$

Claim 1c. Show that K_m for $m \mid n$ are the only subgroups of μ_n

Proof. By a previous homework, we know that subgroups of cyclic groups are cyclic as well. Let G be a cyclic subgroup group of μ_n , $\frac{|\mu_n|}{|G|} \in \mathbb{Z}$. Furthermore since K_m is a cyclic group of order m , $K_m \cong \mu_m$. Thus any G is also isomorphic to some μ_m where $m \mid n$. Thus $K_m = G$. Thus K_m for $m \mid n$ are the only subgroups of μ_n . $\square$

Claim 2. Show every subgroup of the rotations, $C_n = \mu_n$, is normal in D_n .

Proof. For any subgroup G of the rotations $C_n = \mu_n$ by definition G must contain only rotations. $\forall x \in D_n$ and $\forall g \in G$, by the previous homework (Problem 7) xgx^{-1} is a rotation. Furthermore by problem 7, we know if x is a reflection $xgx^{-1} = g^{-1}$ and if $x \in SO(2)$ (or is a rotation) $xgx^{-1} = g$ and since G is a group by definition by the inverse axiom, $g^{-1}, g \in G$. Thus the conjugation of any element of G by any element in D_n gives a element of G , thus $xGx^{-1} = G$ for all $x \in D_n$. Thus G must be normal. And since G is any subgroup of C_n all subgroups of C_n are normal in D_n $\square$

Claim 3a. *Show that when n is odd that all reflections are conjugate in D_n .*

Proof. Let $r_L \in D_n$ be a reflection across the line L . By the definition of D_n , when n is odd, L is a line passing through one vertex, V_i , for some $i \in [0, \dots, n-1]$.

Now, consider any element $g \in D_n$. By a result from a previous problem (problem 7 of the last homework), we know that $gyg^{-1} = r_{g_*(L)}$.

Furthermore, by the definition of the cyclic group C_n , an element $x \in C_n$ represents a rotation that moves a fixed vertex, say V_i , to another vertex, say V_j . Let r_L be the reflection passing through the vertex V_i . Then, for any $x \in C_n$, we have $xr_Lx^{-1} = r_{x_*(L)}$, where $x_*(L)$ is now the image of the line under the rotation x . Importantly, this rotation only changes angles and not distances, so it still passes through V_j .

Thus, $r_{x_*(L)}$ is a reflection in D_n , and since there are exactly n different rotations in C_n , we can obtain all n possible reflections by conjugating them with these rotations.

Therefore, all reflections in D_n are conjugate when n is odd, as they can all be transformed into one another through the conjugation action of the rotations in C_n . $\square$

Claim 3b. *When n is even there are exactly two conjugacy classes – having to do with either vertices, or edges, of the n -gon.*

Proof. Let $r_L \in D_n$ be a reflection across a line L . By the definition of D_n , when n is even, L either passes through 2 vertices and the origin or through the midpoints of 2 edges and the origin.

Now, consider any element $g \in D_n$. By a result from a previous problem (problem 7 of the last homework), we know that $gyg^{-1} = r_{g_*(L)}$.

Furthermore, by the definition of the cyclic group C_n , an element $x \in C_n$ represents a rotation that moves a fixed vertex, say V_i , to another vertex, say V_j .

1. Let r_L be the reflection passing through two vertices V_i and V_b and the origin. Then, for any $x \in C_n$, we have $xr_Lx^{-1} = r_{x_*(L)}$, where $x_*(L)$ is now the image of the line under the rotation x . Importantly, this rotation only changes angles and not distances, so it passes through V_j . Thus, $r_{x_*(L)}$ is a reflection in D_n , and since there are exactly n different rotations in C_n , we can obtain all $n/2$ possible reflections about lines passing through two vertices by conjugating them with these rotations.
2. Let r_L be a reflection across a line passing through the midpoints of 2 edges and the origin, one of the edges is the edge from V_i and $V_{(i+1) \bmod n}$. Then, for any $x \in C_n$, we have $xr_Lx^{-1} = r_{x_*(L)}$, where $x_*(L)$ is now the image of the line under the rotation x . Importantly, this rotation only changes angles and not distances, so it passes through the midpoint of V_j and $V_{(j+1) \bmod n}$. Thus, $r_{x_*(L)}$ is a reflection in D_n , and since there are exactly n different rotations in C_n , we can obtain all $n/2$ possible reflections about lines passing through a midpoint of a edge and the origin by conjugating them with these rotations.

Thus we have n reflections and 2 conjugation classes.

Then, for any $x \in C_n$, we have $xr_Lx^{-1} = r_{x_*(L)}$, where $x_*(L)$ is now the image of the line under the rotation x . Importantly, this rotation only changes angles and not distances, so it still passes through V_j . $\square$

Claim 4. *Show that if n is odd show that the subgroups of C_n are the only normal subgroups (other than D_n itself).*

Proof. Suppose n is odd. Assume the contrary, that $K \subset D_n$ be a normal subgroup of D_n but K is not a subgroup of C_n and $K \neq D_n$. Thus there exists a $r_L \in K$ or a reflection across a line. Let $x \in C_n$ be the generator of C_n . By 3a, we know that $xr_Lx^{-1} \in D_n$ furthermore since K is normal $xr_Lx^{-1} \in K$. Additionally, $xr_Lx^{-1} = r_{x_*(L)}$ is a reflection by problem 7 of the previous homework. Additionally because K is a group and is closed, then $r_Lr_{x_*(L)} \in K$ and $r_Lr_{x_*(L)}$ is a rotation, because $\det(r_Lr_{x_*(L)}) = \det(r_L)\det(r_{x_*(L)}) = -1(-1) = 1$. Since $r_Lr_{x_*(L)}$ is a rotation its in $r_Lr_{x_*(L)} \in C_n$. Furthermore since C_n is a cyclic group $r_Lr_{x_*(L)} = x^k$ for $k \in [0, \dots, n-1]$. Furthermore, $r_Lr_{x_*(L)} = r_Lxr_Lx^{-1}$. By the previous homework, we can represent $r_Lr_{x_*(L)} = r_L^a x^b$ using our relations. $r_Lxr_Lx^{-1} = r_L^2 x^{-1} x^{-1} = x^{-2} = x^{n-2}$. Since n is odd, $n = 2k+1$ for $k \in \mathbb{Z}$, $\gcd(n-2, n) = \gcd(2k-1, 2k+1)$, by Euclid's algorithm $\gcd(2k-1, 2k+1) = \gcd(2k-1, 2k+1 - (2k-1)) = \gcd(2k-1, 2) = 1$. Thus $\gcd(n-2, n) = 1$ and $n-2$ and n are coprime. Thus x^{n-2} generates C_n by beginning group theory 1 problem 6. Thus $r_Lr_{x_*(L)} = x^{n-2}$ is a generator and in K . By Dihedral group 2 problem 3, a generator of C_n and a reflection in D_n following the rules of D_n generate D_n . Thus since K is a subgroups of D_n , D_n rules are followed. Furthermore, a generator $x^{n-2} \in K$ and a reflection of D_n in K . Thus $K = D_n$, forming a contradiction. Thus the subgroups of C_n are the only normal subgroups (other than D_n itself). $\square$

Claim 5a. Now suppose $n = 2k$. Let $N \neq D_n$ be a normal subgroup containing a reflection. Show that N contains $Q := x^2 = \text{rotation by "2 vertices" (the square of the standard generator of } \mu_n)$.

Proof. Let $N \neq D_n$ be a normal subgroup containing a reflection, r_L . Let x be the standard generator of μ_n and thus C_n . Since N is normal $xr_Lx^{-1} \in N$. By closure, $r_Lxr_Lx^{-1} \in N$. Because $xr_L = r_Lx^{-1}$ in D_n by rules of D_n , $r_Lxr_Lx^{-1} = r_L^2 x^{-2} = x^{-2} \in N$. By definition of groups, since $x^{-2} \in N \implies x^2 = Q \in N$. $\square$

Claim 5b. $N \cong D_k = \langle Q, r_L \mid Q^k = r_L^2 = 1, r_L Q r_L = Q^{-1} \rangle$

Proof. $(x^2)^k = x^{2k} = e$ since x generates C_n and $n = 2k$. Thus $|\langle x^2 \rangle| = k$, thus $\langle x^2 \rangle = C_k$. Furthermore N inherits the rule that r_L^2 and $r_L x^2 r_L = (x^2)^{-1}$. By the previous homework, we proved that a group is determined by its generator and relations (its presentation) and a group with at least these rules will be $|N| \leq 2k$. Thus $N = \langle x^2, r_L \mid (x^2)^k = r_L^2 = 1, r_L x^2 r_L = (x^2)^{-1} \rangle$. Furthermore, by our rules in N and the ambient group D_{2k} , every element in $g \in N$ can be written as $g = (x^2)^a r_L^b = x^{2a} r_L^b$ where $a \in [0, \dots, k]$, and $b \in [0, 1]$. Thus $|N| = 2k = |D_k|$. $D_k = \langle Q, r_L \mid Q^k = r_L^2 = 1, r_L Q r_L = Q^{-1} \rangle$. Since both N and D_k have the same presentation there is a surjective homomorphism from $N \rightarrow D_k$ and since they have the same order the map is also injective and thus $N \cong D_k$. $\square$

Claim 5c. Conclude there are exactly two proper normal subgroups not contained in C_n .

Proof. In problem 3b, we have proven that in D_n where n is even, there are 2 conjugacy classes having to do with the vertices of the ngon or the edges of the ngon. Thus if N , a normal subgroup not in C_n , has a reflection r_L across a line through a vertex and the origin then elements in its conjugacy class (proven in 3b), which are all the rotations and reflections across a line through a vertex and the origin, would be in N . Similarly if N , a normal subgroup not C_n , has a reflection r'_L across a line through the midpoint of an edge and the origin then elements in its conjugacy class, which are all the rotations and reflections across a line through the midpoint of an edge and the origin (proven in 3b), would be in N . Thus if we have any reflection in normal group N , then we have its entire conjugacy class. And since there are only 2 conjugacy classes there must be only two normal subgroups not contained in C_n . $\square$

Claim 6. Show that the center of D_n is trivial when n is odd, and $\{I, -I\}$ when n is even (here $-I$ is the map from $\mathbb{R}^2$ to $\mathbb{R}^2$ given by this matrix, it sends each vector to its negative, its rotation by 180 degrees).

Proof. D_n by definition has the rule that $xyx^{-1} = yxy = x^{-1}$ where $x^n = y^2 = 1$. If $n = 2$, then $x = x^{-1}$ b/c $x^2 = 1$. Then $yx = xy$. And everything is commutative.

Suppose $n > 2$, Thus $xy = yx^{-1}$. Let $a \in [0, \dots, n-1]$. Left multiplying by x^{a-1} , we see $x^a y = x^{a-1} y x^{-1} = x^{a-2} y x^{-2}$ and thus $x^a y = y x^{-a}$. $x^{-a} = x^{n-a}$. Thus $x^a y = y x^{n-a}$. Suppose g is an element of the center, by the previous homework $g = x^a y^b$ for some $a \in [1, \dots, n-1]$ and $b \in [0, 1]$. Since g in the center, $gx = x^a y^b x = xg = x^{a+1} y^b$ thus $x^a y^b x = x^{a+1} y^b = x^a x y^b$. Thus after left multiplying by x^{-a} , $y^b x = x y^b$. Assume the contrary, $b = 1$. Then

$$\begin{aligned} yx &= xy \\ &= yx^{-1} \\ x &= x^{-1} \\ x^2 &= e \end{aligned}$$

However the order of x is n and $n > 2$, thus $a^2 \neq e$. Thus $b \neq 0$ and $g = x^a$. Since g in the center, $gy = x^a y = yg = yx^a$

$$\begin{aligned} x^a y &= yx^a \\ &= x^{-a} y \\ &= x^{-a} y \\ x^a &= x^{-a} \\ x^{2a} &= e = x^n \end{aligned}$$

By definition of order of group element, $n \mid 2a$. However since $0 \leq a \leq n$, $a = 0$ or $n = 2a$. If n is odd, $n \neq 2a$ and thus only $a = 0$ is true and $x^0 = e$ in the center of D_n . However if n is even, $n = 2a$ can be true thus $x^{n/2}$ in the center and $x^0 = e$ which is by definition in the center because the identity commutes. $\square$

Problem 7. Explain why its reasonable to view $D_1 := \{1, -1\}$ and D_2 as $\mu_2 \times \mu_2$

Proof. $D_1 = \langle x, y \mid x = y^2 = 1, yxy^{-1} = x \rangle$. $1, -1$ also has the same presentation where $x = 1$ and $y = -1$ and $x = y^2 = 1$ and $yxy^{-1} = yxy = 1 = x$. Thus since the two have the same presentation they are isomorphic and thus the same. $D_2 = \{e, \rho_\pi, r_{x=0}, r_{y=0}\}$ which is isomorphic $\mu_2 \times \mu_2 = \{(1, 1), (-1, -1), (1, -1), (-1, 1)\}$ where the first index is what happens to the x and 2nd is what happens to y and thus $\mu_2 \times \mu_2$ performs the same action as D_2 $\square$

Problem 8. Let N be a normal subgroup of finite index in the group G , and H a subgroup of G . Show that $N \cap H$ is a normal subgroup of H whose index divides the index of N in G (Hint, consider the composition $H \subset G \rightarrow G/N$. What is the kernel? What is the image?) .

Claim 8a. Show that $N \cap H$ is a normal subgroup of H .

Proof. Let $N \triangleleft G$ and $H \leq G$. Then $\forall a \in N \cap H$ and for all $h \in H$, since N is normal subgroup $hah^{-1} \in N$ and since H is closed and $a \in H$ (by definition) $hah^{-1} \in H$. Thus by definition of intersection, $hah^{-1} \in N \cap H$. Thus by definition $N \cap H \triangleleft H$. $\square$

Claim 8b. Index of $N \cap H$ divides the index of N in G .

Proof. Since $H \subset G$, there exists a natural injective map $id : H \rightarrow G$. Let $m : G \rightarrow G/N$. Let $f = m \circ id$. Create a natural quotient homomorphism $q : H \rightarrow H/\ker f$. Furthermore by the definition, $\ker id = \{e\}$ thus $\ker f = N \cap H$. Thus if $a \in H$ and $a \in \ker q$ then by definition $a \in \ker f$. Thus for all $a, b \in H$ if $q(a) = q(b)$, $q(ab^{-1}) = e$ then $ab^{-1} \in \ker q$ and $ab^{-1} \in \ker f$ then $f(a) = f(b)$. Thus $f = F \circ q$ where F is a homomorphism from $H/\ker f \rightarrow H/N$. However since $\ker f = H \cap N$ this map is injective because we have already removed the kernel. Assume there was an element $h \in H$, s.t $F(h \ker f) = eN$ (identity coset of N), then because $f = F \circ q$ must factor properly $f(h) = F(q(h)) = eN$ thus $h \in N$ and thus $h \in \ker f$. Thus F is injective. Thus $H/\ker f$ is a subgroup of G/N . Thus $|H/\ker f|$ divides $|H/N|$, by Lagrange's theorem. By definition, $H/\ker f = H/N \cap H$ and $|H/N \cap H|$ is the index of $N \cap H$ and $|G/N|$ is the index of N . Thus index of $N \cap H$ divides the index of N in G . $\square$

Problem 9. Let N be one of the two special normal subgroups of D_{2k} . Show N has index 2, and contains the center of D_{2k} iff k is even.

Proof. ($\implies$): Suppose N has index 2 and contains the center of D_{2k} . By 6, $Z(D_{2k}) = \{1, -1\}$. Let x be the standard generator of C_{2k} . By 5b, $N \cong D_k$. Let $g \in C_k$ be the standard generator of C_k . By 5b $g \cong x^2$. Since $-1 = x^a$ and $-1^2 = 1$ so $x^{2a} = e$. Thus order of x^{2a} is 2. Since $x^{2a} = g^a \in N \cong D_k$, and $g^a \in C_k$. By Lagrange's theorem $|\langle g^a \rangle| \mid |C_k|$ thus $2 \mid k$ and k must be even.

($\impliedby$) We know that $Z(D_{2k}) = 1, -1$, and since k is even, there exists an element in D_{2k} of the form x^k , where x is the standard generator of C_{2k} . This element is equal to -1 in D_{2k} because $x^k \neq e$ but $(x^k)^2 = x^{2k} = e$. Since all the cosets of N have the same order since N is normal, $|D_{2k}| = |N| [D_k : N]$. Thus $[D_k : N] = |D_{2k}|/|N|$. By Lagrange's theorem $|N|$ divides $|D_{2k}|$. By 5b, $N \cong D_k$, thus $|N| = |D_k| = 2k$. Thus $[D_k : N] = 4k/2k = 2$. Thus index of N is 2. $\square$

Claim 10. Let H be a subgroup of D_n , not contained in C_n . Let $R := C_n \cap H$. Explain why this has index 2 in H , and generated by x^m for some m dividing n , where x generates C_n . Show H is isomorphic to D_m , generated by x^k and any of its reflections.

Proof. By the counting formula, we know $[H : R] = |H|/|R|$. $|R|$ will be a subgroup of H consisting of only rotations and $|H|$ subgroup of C_n because all rotations are in C_n and $\forall r_a, r_b \in R$ $r_a, r_b \in H$ and by closure $r_a r_b \in H$ and $r_a r_b \in C_n$ and $r_a r_a b \in R$. Furthermore since C_n and H are groups they must maintain properties of inverses and identity so $C_n \cap H$ must have inverses and identities as well for its common elements. By a previous homework, we know $C_n \cong \mu_n$. By problem 1, we know, K_m for $m \mid n$ are the only subgroups of μ_n . Thus $R \cong K_m$. Thus $|R| = m$. Furthermore since $H \not\subset C_n$, there exists a reflection $y \in H$. Since generator of $K_m \cong R \in H$ and $y \in H$ and $x^m = e = y^2$, where x is the standard generator of $K_m \cong R$ and y is a reflection. Furthermore $ygy^{-1} = g^{-1}$ is true (g generator of C_n). Then $yg = g^{-1}y$ is true. Then $yg^a = g^{-1}y$. Since $x \in \langle g \rangle$, $yx = x^{-1}y$ is true and $xyy^{-1} = x^{-1}$ in H . Thus $H = \langle x, y \mid x^m = y^2 = 1, yxy^{-1} = x^{-1} \rangle$ and $|H| \leq 2m$. Furthermore by the

previous homework, any group with at least these rules will have an expression for each of its elements of the form $x^a y^b$ where $a \in [0, \dots, m]$ and $b \in [0, 1]$. Thus $|H| = 2m = |D_m|$. Thus since a group is determined by its generator and relations there exists a surjective homomorphism from this group to other groups with the same generators and at least these rules. Furthermore since the orders of both groups are equal, the homomorphism is also injective thus there exists an isomorphism from $H \rightarrow D_m$ and $H \cong D_m$. Thus $|H| = 2m$. And $[H : R] = 2m/m = 2$. $\square$